1 A Data Odyssey: Relational DBMS vs. GGG



Data Rep. in RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep. in GGG

Real World Entity

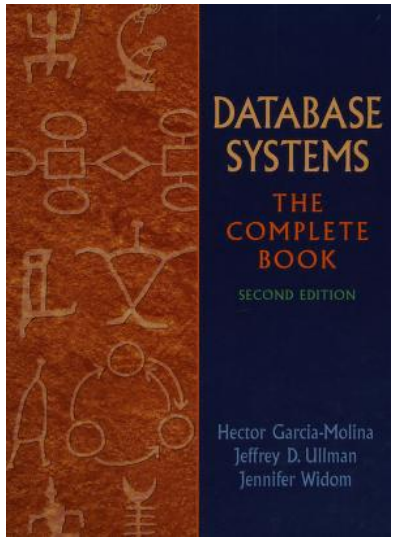
Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

Outline: W07: Physical Modeling: SQL



CH06 (2nd Ed.)

I) Introduction

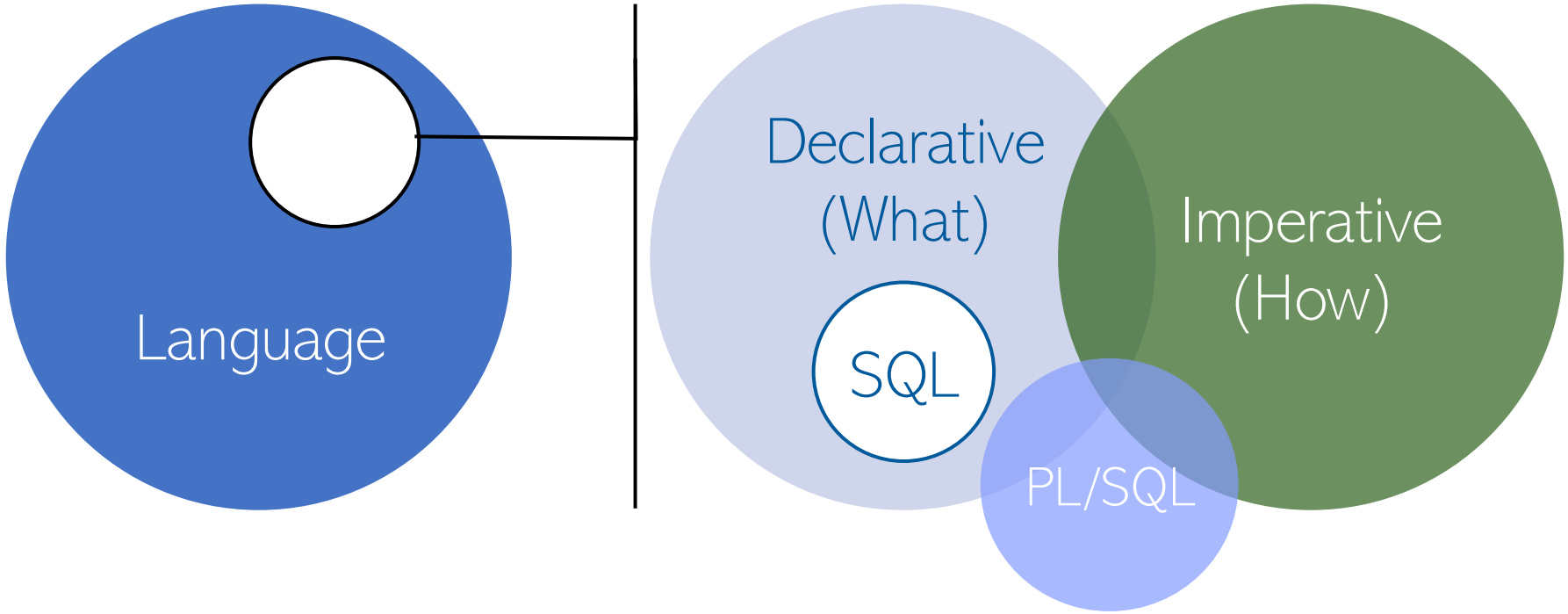
II) Data Definition Language (DDL)

III) Data Modification Language (DML)

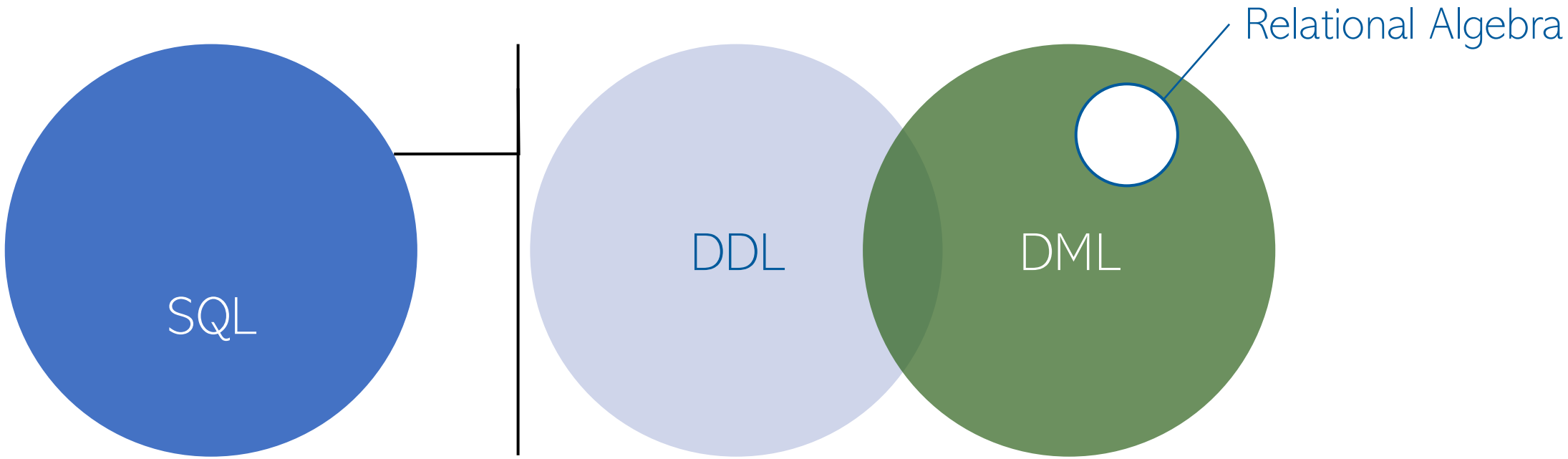
IIII) Data Query Language (`SELECT`)

Hands-On

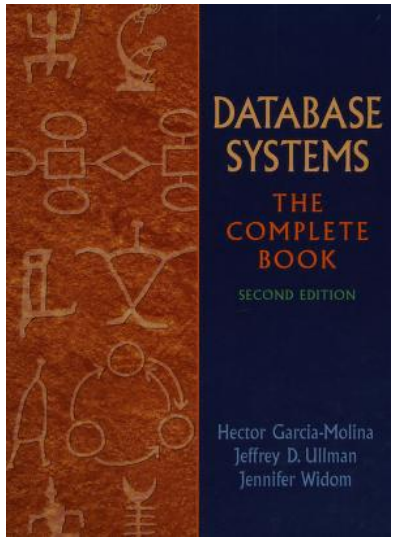
Programming Languages: SQL is Declarative



4 SQL: DDL + DML



Outline: W07: Physical Modeling: SQL



CH06 (2nd Ed.)

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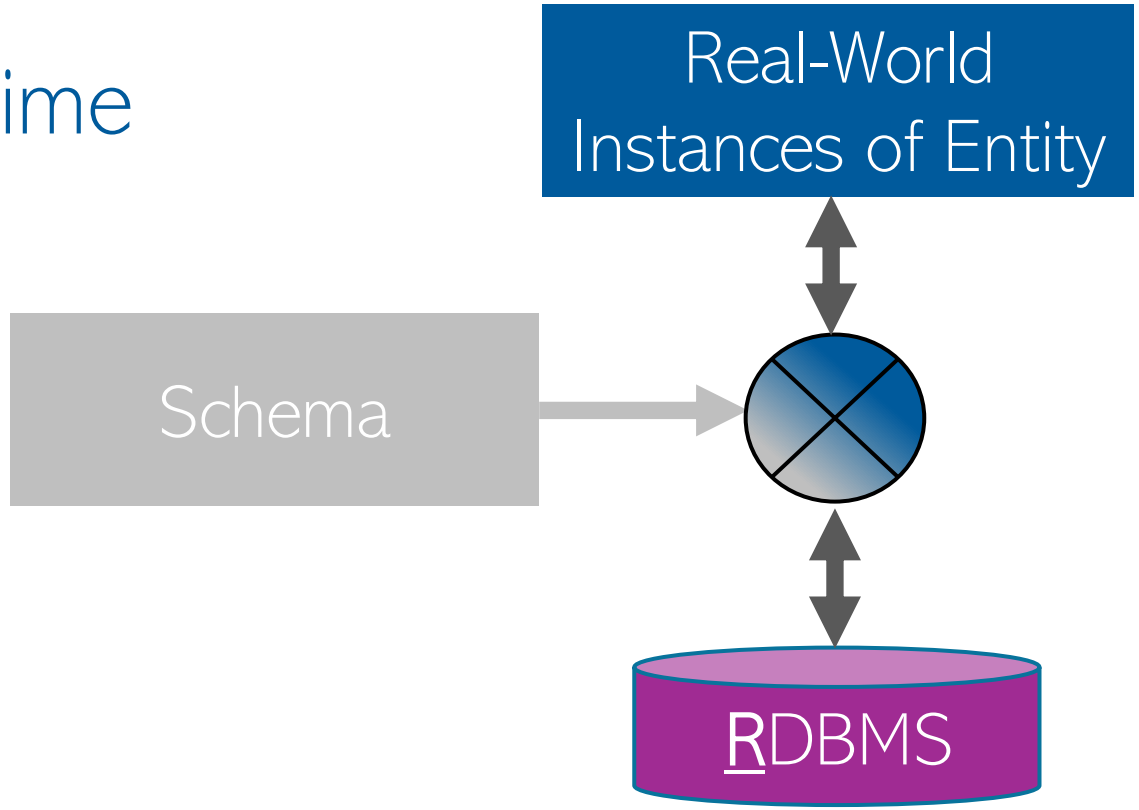
III) Data Modification Language (DML)

IIII) Data Query Language (`SELECT`)

Hands-On

SQL: DML for Entity Instances (Tuples)

Actual Entities (Tuple)
Rapidly Change at Running Time
Must Conform to Schema

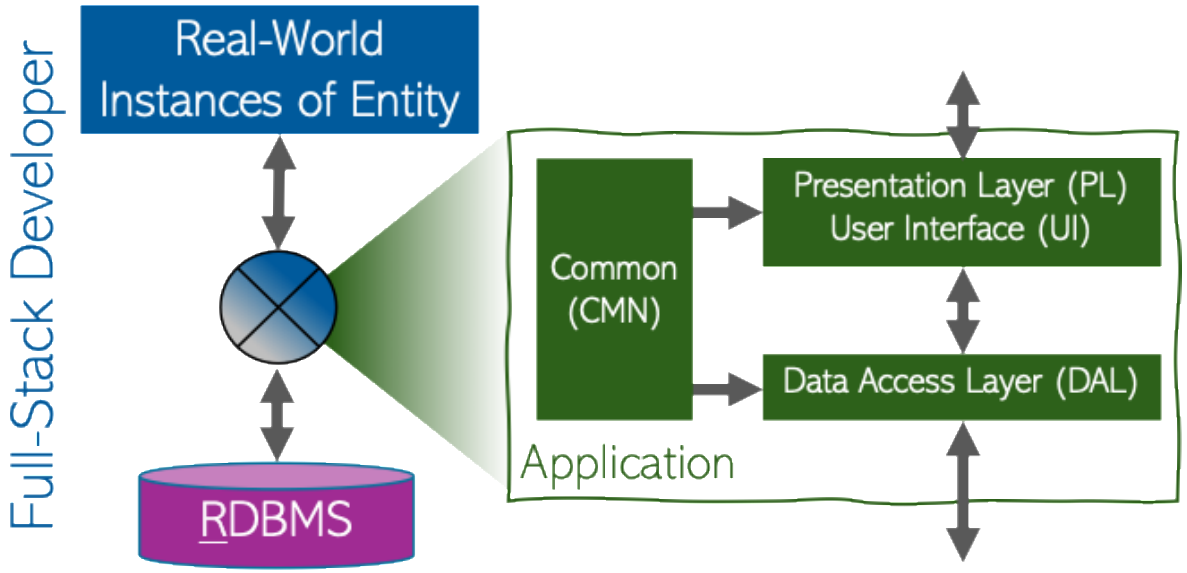


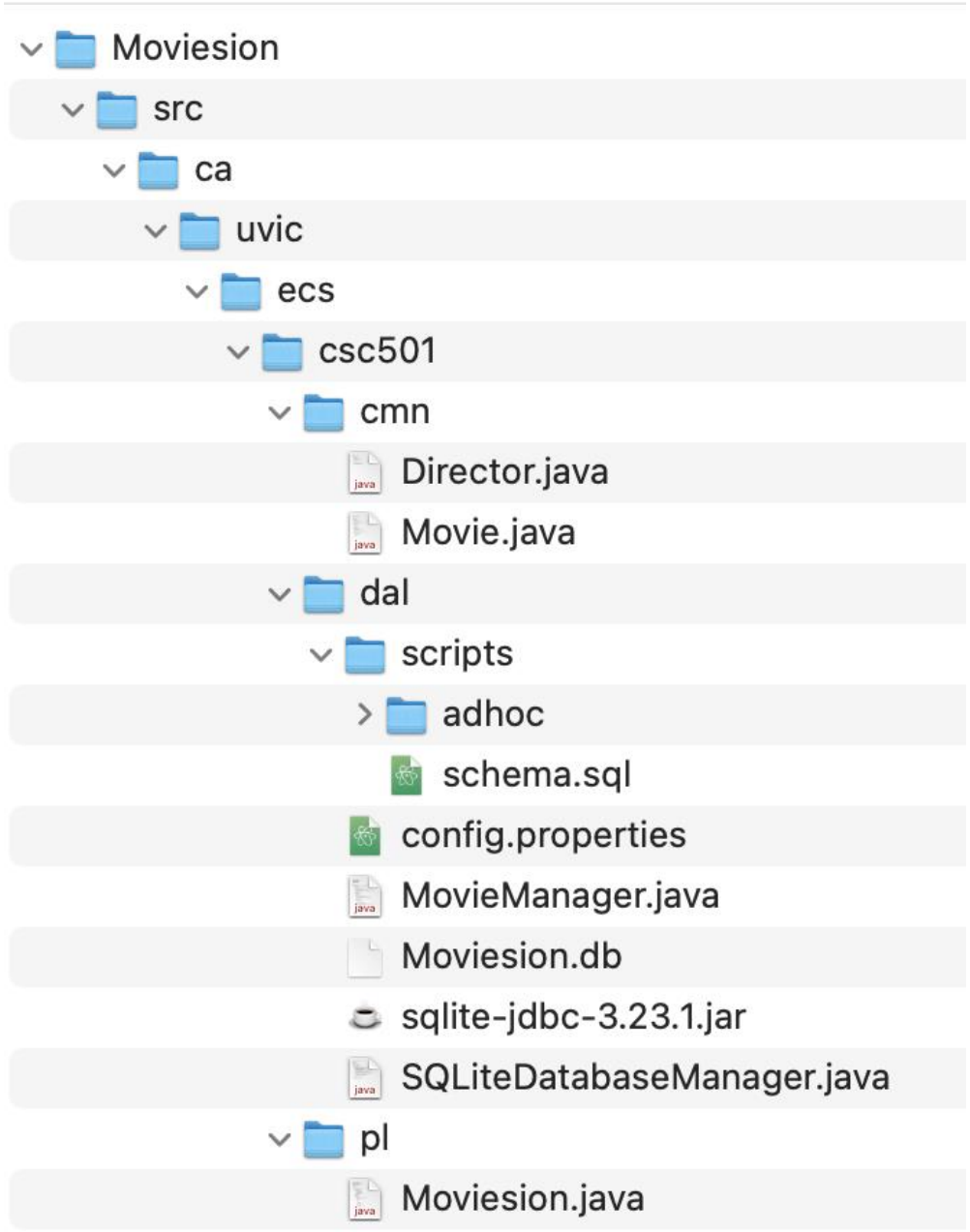
→ Data Access Layer (DAL)



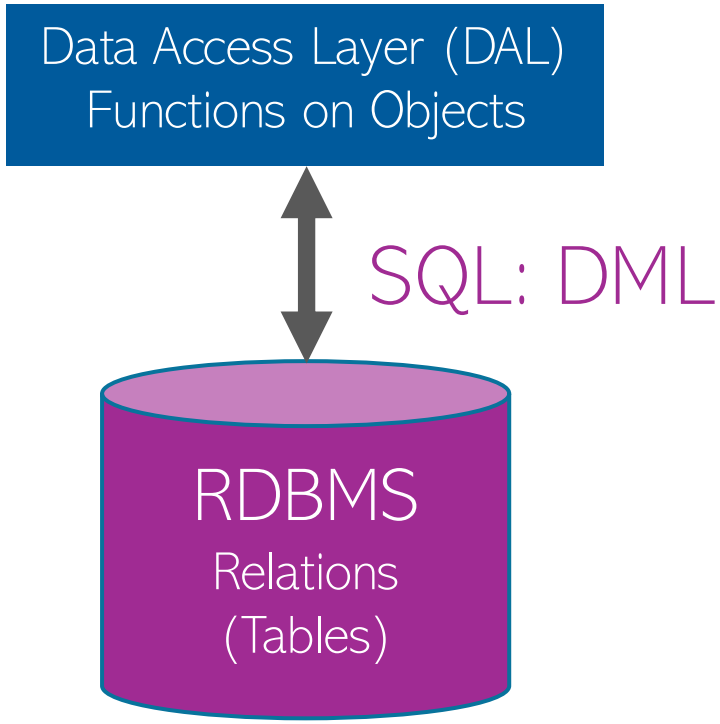
The Matrix (1999), He is the One.

SQL: DML via DAL





SQL: DML via DAL by Object Relational Mapping (ORM)



Object

Relation (Table)

```
public void add(Movie newMovie){
    ...
    String sql = "INSERT INTO Director(Name) VALUES(?)";
    ps = db.prepareStatement(...);
    ps.setString(1, newMovie.getName());
    rowsAffected = ps.executeUpdate();
    ...
}
```

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SQL: DML

Data Manipulation Language to

```
INSERT
UPDATE
DELETE
SELECT
```

to/from relations (tables).

SQL: DML: INSERT

```
INSERT INTO TableName(c1, c2, ....) VALUES (v1, v2, ...);
```

- The number of columns and values must be same.
- If column list omitted, the order in original table is assumed.
- Data types of values must be compatible with the respective columns.

```
INSERT INTO Director(FirstName, LastName) VALUES ('Alfred', 'Hitchcock');
INSERT INTO Director VALUES ('Alfred', 'Hitchcock');
INSERT INTO Director(FirstName, LastName) VALUES (2, 'Hitchcock');
```

SQL: DML: UPDATE

```
UPDATE TableName SET c1=v1, c2=v2, ... [WHERE θ];
```

- Data types of values must be with the respective columns.
- Only rows which satisfy the θ condition will be updated.
- If there is no θ, all rows will be updated!

```
UPDATE Director SET LastName='Hitchkok' WHERE Id = 1;
1 rows affected
UPDATE Director SET LastName='Hitchkok' WHERE LastName = 'Hitchcock';
3 rows affected
UPDATE Director SET LastName='Hitchkok'
36 rows affected
```

SQL: DML: DELETE

DELETE FROM TableName [WHERE θ];

- Only rows which satisfy the θ condition will be deleted.
- If there is no θ, all rows will be deleted.

```
DELETE FROM Director WHERE Id = 1;
1 rows affected
DELETE FROM Director WHERE LastName = 'Hitchcock';
3 rows affected
DELETE FROM Director
36 rows affected
```

SQL: DML: Data Integrity

DML manipulated tuples in a database. So, after each operation, DBMS checks the integrity of database:

I. Domain Integrity (Data Type Integrity)
II. Entity Integrity
III. Referential Integrity
IIII. User-defined Integrity

DML will fail if they violate (conflict with) any integrity constraints!

SQL: DML: Data Integrity

DML manipulated tuples in a database. So, after each operation, DBMS checks the integrity of database:

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DML manipulated tuples in a database. So, after each operation, DBMS checks the integrity of database:

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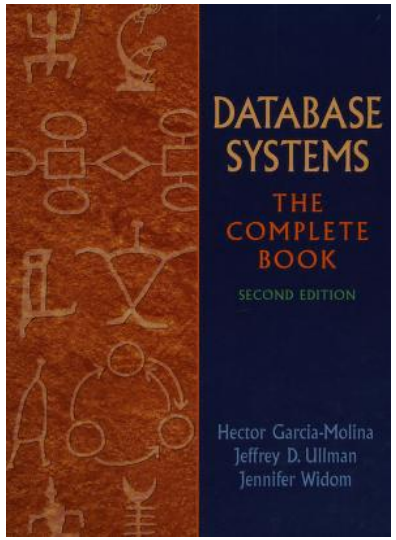
DML will fail if they violate (conflict with) any integrity constraints!

SQL: DML: Data Integrity: User-defined Integrity

- Any constraints that are missing in ER and Relational modeling, can be defined using `CHECK`.

- However, few designer does that. Why?
- Few designers even disable data integrity. Why?

Outline: W07: Physical Modeling: SQL



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III) Data Modification Language (DML)

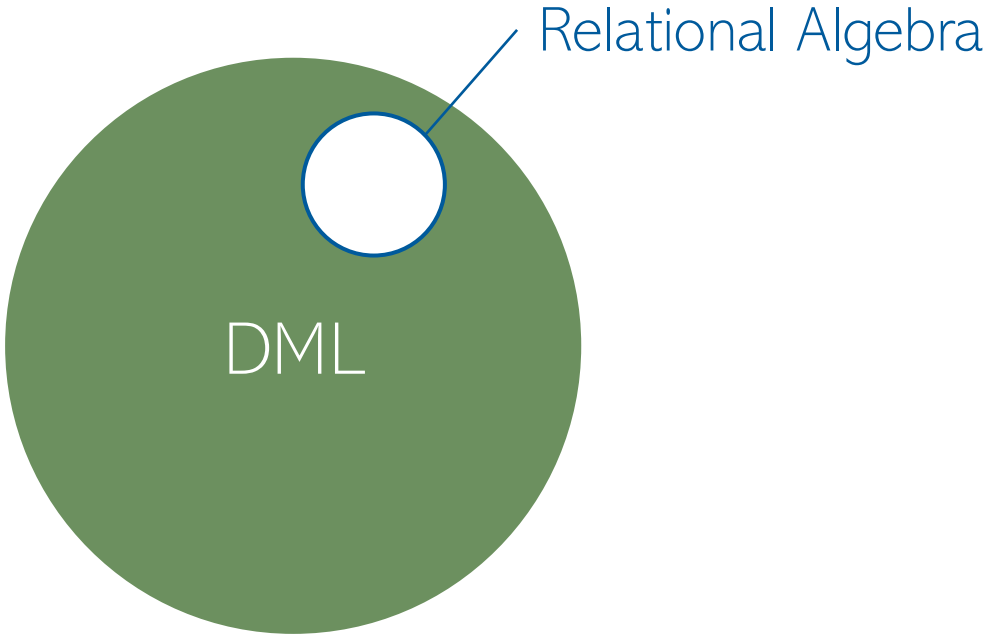
IIII) Data Query Language (`SELECT`)

Hands-On

Outline: W07: Physical Modeling: SQL

Relational algebra , i.e.,

5 `SELECT DISTINCT` Columns
1 `FROM` Tables
2 [`WHERE` θ]
3 [`GROUP BY` Columns]
4 [`HAVING` θ′]
6 [`ORDER BY` Columns [`ASC` | `DESC`]]
7 [`LIMIT` # [`OFFSET` #]]



SQL: Projection (π)

What are directors' names and their birth places?

$\pi_{\text{FirstName, LastName, MovieCount}}(\text{Director})$

```
5 SELECT FirstName, LastName, MovieCount
1 FROM Director
```

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803

SQL: Projection (π) by Aggregation Functions

`COUNT`, `SUM`, `MIN`, `MAX`, `AVG`

How many directors, and the birth date of the oldest director?

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803

5 `SELECT COUNT`(Id) `AS` Count, `MIN`(BirthDate) `AS` Oldest
1 `FROM` Director

Count	Oldest
3	Aug. 13, 1899

23 SQL: Selection (σ)

Who are American directors?

$$\sigma_{\text{BirthPlace}=\text{'USA'}}(\text{Director})$$

```
3 SELECT Id, FirstName, LastName, BirthDate, BirthPlace, BestMovieId
1 FROM Director
2 WHERE BirthPlace= 'USA'
```

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803

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SQL: Selection (∫) by Aggregation Functions

COUNT, SUM, MIN, MAX, AVG

How many American directors?

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803

3 `SELECT COUNT(Id) AS AmericanDirectorsCount`
1 `FROM Director`
2 `WHERE BirthPlace='USA'`

AmericanDirectorsCount
2

25 SQL: Selection (∫) then GROUP BY then Aggregation

What are countries with at least two directors in 20th century?

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803
4	Roman	Polanski	Aug. 18, 1933	France	871

```
5 SELECT BirthPlace AS Country, COUNT(Id) AS DirectorsCount
1 FROM Director
2 WHERE BirthDate>=1900
3 GROUP BY BirthPlace
4 HAVING COUNT(Id) >= 2
```

Country	DirectorsCount
USA	2

26 SQL: Sorting

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803
4	Roman	Polanski	Aug. 18, 1933	France	871

5 1 6

```
SELECT Id, FirstName, LastName, BirthDate, BirthPlace
FROM Director
ORDER BY BirthPlace, BirthDate DESC
```

Director				
Id	FirstName	LastName	BirthDate	BirthPlace
2	Alfred	Hitchcock	Aug. 13, 1899	England
4	Roman	Polanski	Aug. 18, 1933	France
3	Clint	Eastwood	May 31, 1930	USA
1	Stanley	Kubrick	Jul. 26, 1928	USA

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SQL: Paging

Top-N tuples After skipping Top-M ones

5 `SELECT` FirstName, LastName
1 `FROM` Director
6 `ORDER BY` LastName
7 `LIMIT` 10 `OFFSET` 2×10

FirstName	LastName
Pedro	Almodóvar
Wes	Anderson
Darren	Aronofsky
Paul Thomas	Anderson
Robert	Altman
Ingmar	Bergman
Kathryn	Bigelow
James	Cameron
Sofia	Coppola
Francis Ford	Coppola
Joel	Coen
Ethan	Coen
Alfonso	Cuarón
Guillermo	del Toro
Lana	Wachowski
Lilly	Wachowski
Claire	Denis
Céline	Sciamma
David	Fincher
John	Ford
Jean-Luc	Godard
Greta	Gerwig
Werner	Herzog
Alfred	Hitchcock
Peter	Jackson
Bong	Joon-ho
Alejandro	González Iñárritu
Stanley	Kubrick
Akira	Kurosawa
Ang	Lee
Spike	Lee
Richard	Linklater
George	Lucas
Terrence	Malick
Christopher	Nolan
Jordan	Peele
Roman	Polanski
Satyajit	Ray
Martin	Scorsese
Steven	Spielberg
Quentin	Tarantino
Andrei	Tarkovsky
Denis	Villeneuve
Lars	von Trier
Orson	Welles
Chloé	Zhao

Skipping Top-M=2×N
2 pages of N tuples

Top-N tuples
1 page of N tuples
3rd page

28

SQL: Paging

Who is the oldest director?

Director					
Id	FirstName	LastName	BirthDate	BirthPlace	BestMovieId
1	Stanley	Kubrick	Jul. 26, 1928	USA	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	203
3	Clint	Eastwood	May 31, 1930	USA	803

5 `SELECT` FirstName, LastName, BirthDate, BirthPlace
1 `FROM` Director
6 `ORDER BY` BirthDate
7 `LIMIT` 1



FirstName	LastName	BirthDate	BirthPlace
Alfred	Hitchcock	Aug. 13, 1899	England
Stanley	Kubrick	Jul. 26, 1928	USA
Clint	Eastwood	May 31, 1930	USA

29 SQL: Cartesian Product (×)

5 `SELECT DISTINCT` Columns
1 `FROM` Tables → Table1, Table2, …, TableN
2 [`WHERE` θ]
3 [`GROUP BY` Columns]
4 [`HAVING` θ′]
6 [`ORDER BY` Columns [`ASC` | `DESC`]]
7 [`LIMIT` N [`OFFSET` M]]

SQL: Cartesian Product (×)

$R1(a_1, a_2, \ldots, a_n)$
$R2(b_1, b_2, \ldots, b_m)$

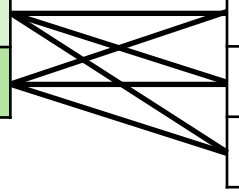
$R = R1 \times R2$ is a relation with:

- All attributes in R1 and R2: $R(a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_m)$
- Each tuple of R1 is paired with all tuples of R2

SQL: Cartesian Product (×)

R1 × R2 = R

a_1	a_2	…	a_n
A1	A2	A3	A4
B1	B2	B3	B4



b_1	b_2	…	b_m
C1	C2	C3	C4
D1	D2	D3	D4
E1	E2	E3	E4

a_1	a_2	…	a_n	b_1	b_2	…	b_m
A1	A2	A3	A4	C1	C2	C3	C4
A1	A2	A3	A4	D1	D2	D3	D4
A1	A2	A3	A4	E1	E2	E3	E4
B1	B2	B3	B4	C1	C2	C3	C4
B1	B2	B3	B4	D1	D2	D3	D4
B1	B2	B3	B4	E1	E2	E3	E4

SQL: Cartesian Product (×): Pairing Operation

R1 and R2 can be any relations (tables). No restrictions.

Any two relations can be paired!
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Any two relations can be paired!

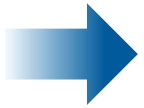
But, which relations are **more interesting** for pairing?

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie and User?

Movie			
Id	Title	Language	RunningTime
1	2001: A Space Odyssey	English	142
2	Rosemary's Baby	English	136

User		
Id	Username	Password
1	fani	***
2	cjason	***
3	h_f492	***



Id	Title	Language	RunningTime	Id	Username	Password
1	2001: A Space Odyssey	English	142	1	fani	***
1	2001: A Space Odyssey	English	142	2	cjason	***
1	2001: A Space Odyssey	English	142	3	h_f492	***
2	Rosemary's Baby	English	136	1	fani	***
2	Rosemary's Baby	English	136	2	cjason	***
2	Rosemary's Baby	English	136	3	h_f492	***



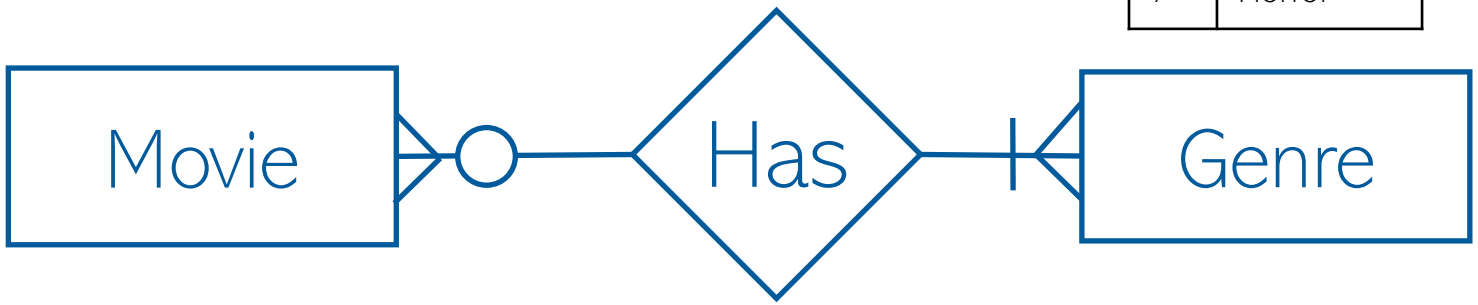
SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Movie			
Id	Title	Language	RunningTime
1	2001: A Space Odyssey	English	142
2	Rosemary's Baby	English	136

MovieGenre	
MovieId	GenreId
1	1
1	3
2	6
2	7
2	2

Genre	
Id	Title
1	Sci-fi
2	Action
3	Adventure
4	Comedy
5	Crime
6	Drama
7	Horror

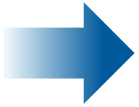


SQL: Cartesian Product (×): Pairing Operation

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Id	Title	Language	RunningTime
1	2001: A Space Odyssey	English	142
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MovieGenre	
MovieId	GenreId
1	1
1	3
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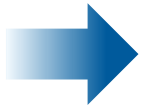
Movie × MovieGenre					
Id	Title	Language	RunningTime	MovieId	GenreId
1	2001: A Space Odyssey	English	142	1	1
1	2001: A Space Odyssey	English	142	1	3
1	2001: A Space Odyssey	English	142	2	6
1	2001: A Space Odyssey	English	142	2	7
1	2001: A Space Odyssey	English	142	2	2
2	Rosemary's Baby	English	136	1	1
2	Rosemary's Baby	English	136	1	3
2	Rosemary's Baby	English	136	2	6
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SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

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Id	Title	Language	RunningTime
1	2001: A Space Odyssey	English	142
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MovieGenre	
MovieId	GenreId
1	1
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2	6
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Movie × MovieGenre					
Id	Title	Language	RunningTime	MovieId	GenreId
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1	2001: A Space Odyssey	English	142	2	7
1	2001: A Space Odyssey	English	142	2	2
2	Rosemary's Baby	English	136	1	1
2	Rosemary's Baby	English	136	1	3
2	Rosemary's Baby	English	136	2	6
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SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Id	Title	Language	RunningTime	MovieId	GenreId
1	2001: A Space Odyssey	English	142	1	1
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2	Rosemary's Baby	English	136	2	6
2	Rosemary's Baby	English	136	2	7
2	Rosemary's Baby	English	136	2	2

$$\int_{\text{Movie.Id = MovieGenre.MovieId}} (\text{Movie} \times \text{MovieGenre})$$

```
SELECT ...
FROM Movie, MovieGenre
WHERE  Movie.Id = MovieGenre.MovieId
```

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Id	Title	Language	RunningTime	MovieId	GenreId
1	2001: A Space Odyssey	English	142	1	1
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2	Rosemary's Baby	English	136	2	7
2	Rosemary's Baby	English	136	2	2

$$\int_{\text{Movie.Id = MovieGenre.MovieId}} (\text{Movie} \times \text{MovieGenre})$$

```
SELECT ...
FROM Movie
INNER JOIN MovieGenre ON Movie.Id = MovieGenre.MovieId
```

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Movie × MovieGenre					
Id	Title	Language	RunningTime	MovieId	GenreId
1	2001: A Space Odyssey	English	142	1	1
1	2001: A Space Odyssey	English	142	1	3
2	Rosemary's Baby	English	136	2	6
2	Rosemary's Baby	English	136	2	7
2	Rosemary's Baby	English	136	2	2

Genre	
Id	Title
1	Sci-fi
2	Action
3	Adventure
4	Comedy
5	Crime
6	Drama
7	Horror



Id	Title	Language	RunningTime	MovieId	GenreId	Id	Title
1	2001: A Space Odyssey	English	142	1	1	1	Sci-fi
1	2001: A Space Odyssey	English	142	1	1	2	Action
1	2001: A Space Odyssey	English	142	1	1	3	Adventure
1	2001: A Space Odyssey	English	142	1	1	4	Comedy
1	2001: A Space Odyssey	English	142	1	1	5	Crime
1	2001: A Space Odyssey	English	142	1	1	6	Drama
1	2001: A Space Odyssey	English	142	1	1	7	Horror
1	2001: A Space Odyssey	English	142	1	3	1	Sci-fi
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1	2001: A Space Odyssey	English	142	1	3	3	Adventure
1	2001: A Space Odyssey	English	142	1	3	4	Comedy
1	2001: A Space Odyssey	English	142	1	3	5	Crime
1	2001: A Space Odyssey	English	142	1	3	6	Drama
1	2001: A Space Odyssey	English	142	1	3	7	Horror
2	Rosemary's Baby	English	136	2	6	1	Sci-fi
2	Rosemary's Baby	English	136	2	6	2	Action
2	Rosemary's Baby	English	136	2	6	3	Adventure
2	Rosemary's Baby	English	136	2	6	4	Comedy
2	Rosemary's Baby	English	136	2	6	5	Crime
2	Rosemary's Baby	English	136	2	6	6	Drama
2	Rosemary's Baby	English	136	2	6	7	Horror
2	Rosemary's Baby	English	136	2	7	1	Sci-fi
2	Rosemary's Baby	English	136	2	7	2	Action
2	Rosemary's Baby	English	136	2	7	3	Adventure
2	Rosemary's Baby	English	136	2	7	4	Comedy
2	Rosemary's Baby	English	136	2	7	5	Crime
2	Rosemary's Baby	English	136	2	7	6	Drama
2	Rosemary's Baby	English	136	2	7	7	Horror
2	Rosemary's Baby	English	136	2	2	1	Sci-fi
2	Rosemary's Baby	English	136	2	2	2	Action
2	Rosemary's Baby	English	136	2	2	3	Adventure
2	Rosemary's Baby	English	136	2	2	4	Comedy
2	Rosemary's Baby	English	136	2	2	5	Crime
2	Rosemary's Baby	English	136	2	2	6	Drama
2	Rosemary's Baby	English	136	2	2	7	Horror

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Movie × MovieGenre					
Id	Title	Language	RunningTime	MovieId	GenreId
1	2001: A Space Odyssey	English	142	1	1
1	2001: A Space Odyssey	English	142	1	3
2	Rosemary's Baby	English	136	2	6
2	Rosemary's Baby	English	136	2	7
2	Rosemary's Baby	English	136	2	2

Genre	
Id	Title
1	Sci-fi
2	Action
3	Adventure
4	Comedy
5	Crime
6	Drama
7	Horror

Id	Title	Language	RunningTime	MovieId	GenreId	Id	Title
1	2001: A Space Odyssey	English	142	1	1	1	Sci-fi
1	2001: A Space Odyssey	English	142	1	1	2	Action
1	2001: A Space Odyssey	English	142	1	1	3	Adventure
1	2001: A Space Odyssey	English	142	1	1	4	Comedy
1	2001: A Space Odyssey	English	142	1	1	5	Crime
1	2001: A Space Odyssey	English	142	1	1	6	Drama
1	2001: A Space Odyssey	English	142	1	1	7	Horror
1	2001: A Space Odyssey	English	142	1	3	1	Sci-fi
1	2001: A Space Odyssey	English	142	1	3	2	Action
1	2001: A Space Odyssey	English	142	1	3	3	Adventure
1	2001: A Space Odyssey	English	142	1	3	4	Comedy
1	2001: A Space Odyssey	English	142	1	3	5	Crime
1	2001: A Space Odyssey	English	142	1	3	6	Drama
1	2001: A Space Odyssey	English	142	1	3	7	Horror

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Id	Title	Language	RunningTime	MovieId	GenreId	Id	Title
1	2001: A Space Odyssey	English	142	1	1	1	Sci-fi
1	2001: A Space Odyssey	English	142	1	3	3	Adventure

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Movie × MovieGenre					
Id	Title	Language	RunningTime	MovieId	GenreId
1	2001: A Space Odyssey	English	142	1	1
1	2001: A Space Odyssey	English	142	1	3
2	Rosemary's Baby	English	136	2	6
2	Rosemary's Baby	English	136	2	7
2	Rosemary's Baby	English	136	2	2

Genre	
Id	Title
1	Sci-fi
2	Action
3	Adventure
4	Comedy
5	Crime
6	Drama
7	Horror

Id	Title	Language	RunningTime	MovieId	GenreId	Id	Title
2	Rosemary's Baby	English	136	2	6	1	*Sci-fi*
2	Rosemary's Baby	English	136	2	6	2	*Action*
2	Rosemary's Baby	English	136	2	6	3	*Adventure*
2	Rosemary's Baby	English	136	2	6	4	*Comedy*
2	Rosemary's Baby	English	136	2	6	5	*Crime*
2	Rosemary's Baby	English	136	2	6	6	*Drama*
2	Rosemary's Baby	English	136	2	6	7	*Horror*
2	Rosemary's Baby	English	136	2	7	1	*Sci-fi*
2	Rosemary's Baby	English	136	2	7	2	*Action*
2	Rosemary's Baby	English	136	2	7	3	*Adventure*
2	Rosemary's Baby	English	136	2	7	4	*Comedy*
2	Rosemary's Baby	English	136	2	7	5	*Crime*
2	Rosemary's Baby	English	136	2	7	6	*Drama*
2	Rosemary's Baby	English	136	2	7	7	*Horror*
2	Rosemary's Baby	English	136	2	2	1	*Sci-fi*
2	Rosemary's Baby	English	136	2	2	2	*Action*
2	Rosemary's Baby	English	136	2	2	3	*Adventure*
2	Rosemary's Baby	English	136	2	2	4	*Comedy*
2	Rosemary's Baby	English	136	2	2	5	*Crime*
2	Rosemary's Baby	English	136	2	2	6	*Drama*
2	Rosemary's Baby	English	136	2	2	7	*Horror*

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

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1	2001: A Space Odyssey	English	142	1	3	3	Adventure
2	Rosemary's Baby	English	136	2	6	6	Drama
2	Rosemary's Baby	English	136	2	7	7	Horror
2	Rosemary's Baby	English	136	2	2	2	Action

$$\int_{\text{Movie.Id = MovieGenre.MovieId \& MovieGenre.GenreId = Genre.Id}} (\text{Movie} \times \text{MovieGenre} \times \text{Genre})$$

```
SELECT ...
FROM Movie, MovieGenre, Genre
WHERE  Movie.Id = MovieGenre.MovieId AND  MovieGenre.GenreId = Genre.Id
```

SQL: Cartesian Product (×): Pairing Operation

Any meaningful information by pairing Movie, Genre, MovieGenre?

Id	Title	Language	RunningTime	MovieId	GenreId	Id	Title
1	2001: A Space Odyssey	English	142	1	1	1	Sci-fi
1	2001: A Space Odyssey	English	142	1	3	3	Adventure
2	Rosemary's Baby	English	136	2	6	6	Drama
2	Rosemary's Baby	English	136	2	7	7	Horror
2	Rosemary's Baby	English	136	2	2	2	Action

$$\int_{\text{Movie.Id = MovieGenre.MovieId \& MovieGenre.GenreId = Genre.Id}} (\text{Movie} \times \text{MovieGenre} \times \text{Genre})$$

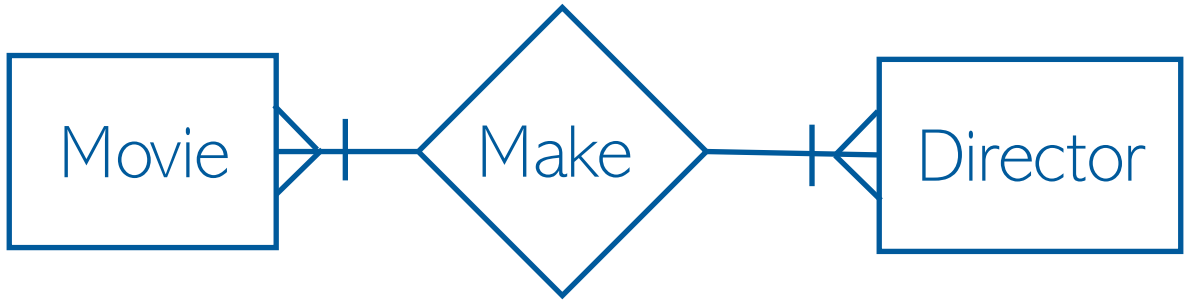
```
SELECT ...
FROM Movie
INNER JOIN MovieGenre ON Movie.Id = MovieGenre.MovieId
INNER JOIN Genre      ON MovieGenre.GenreId = Genre.Id
```

SQL: Cartesian Product (×): Pairing Operation

While relations (tables) can be paired without restrictions, but

- Those have relationship in ER diagram
- Those have linked by Foreign Keys (FK) in relational model

SQL: Cartesian Product (×): Pairing Operation

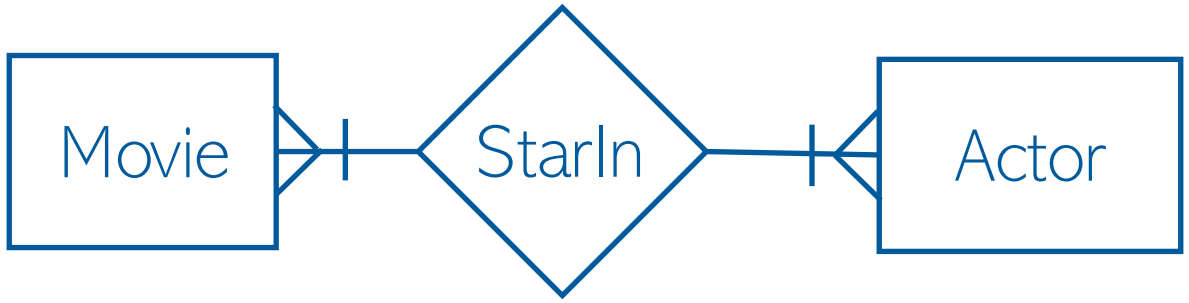


Who made 'Pulp Fiction'?

```
SELECT ...
FROM Movie, MovieDirector, Director
WHERE  Movie.Title = 'Pulp Fiction'
    AND Movie.Id = MovieDirector.MovieId
    AND MovieDirector.DirectorId = Director.Id
```

```
SELECT ...
FROM Movie
WHERE  Movie.Title = 'Pulp Fiction'
INNER JOIN MovieDirector ON Movie.Id = MovieDirector.MovieId
INNER JOIN Director ON MovieDirector.DirectorId = Director.Id
```

SQL: Cartesian Product (×): Pairing Operation



Who acted in 'Pulp Fiction'?

```
SELECT ...
FROM Movie, MovieActor, Actor
WHERE  Movie.Title = 'Pulp Fiction'
    AND Movie.Id = MovieActor.MovieId
    AND MovieActor.ActorId = Actor.Id
```

```
SELECT ...
FROM Movie
WHERE  Movie.Title = 'Pulp Fiction'
INNER JOIN MovieActor ON Movie.Id = MovieActor.MovieId
INNER JOIN Actor ON MovieActor.ActorId = Actor.Id
```

SQL: Cartesian Product (×): Pairing Operation

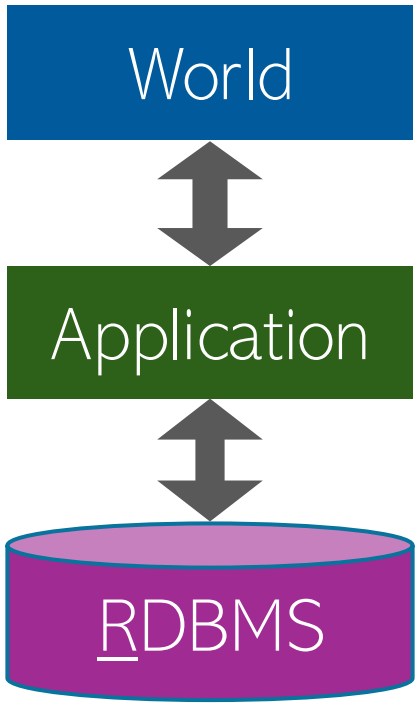
Product is the most important operator in relational model.

Product is the most common operator in relational model.

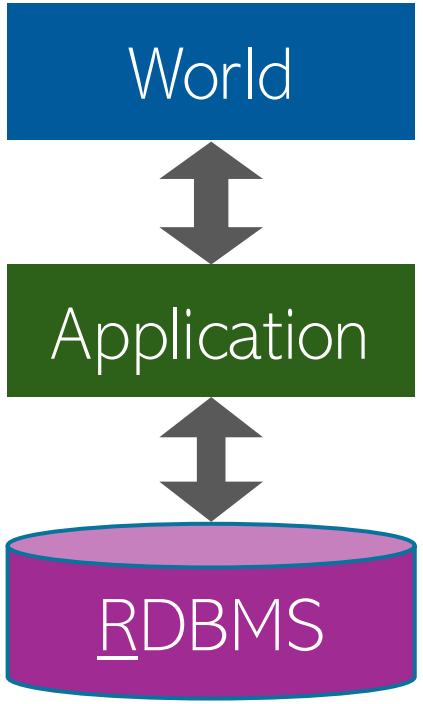
Product is the most expensive operator in relational model!

SQL: DBMS vs. Application Processing

How many movies do we have?



SQL: DBMS vs. Application Processing

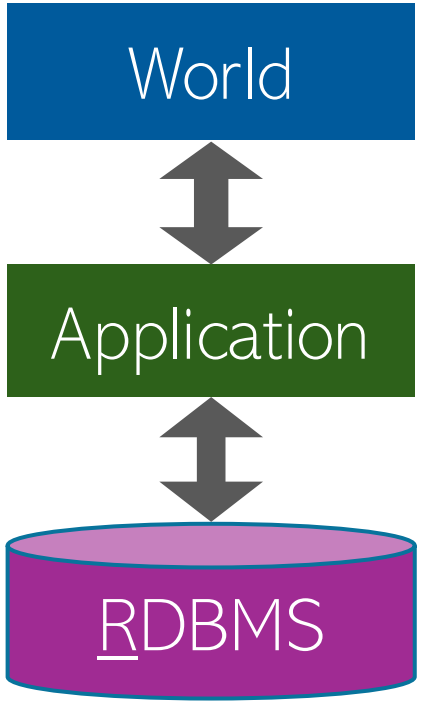


How many movies do we have?

Generate `SELECT` with `COUNT` …

→ Return the count on the Movie's tuples

SQL: DBMS vs. Application Processing



How many movies do we have?

Generate `SELECT` for all tuples

→ Count the returned tuples …

Return Movie's tuples

SQL: DBMS vs. Application Processing

Application Processing

Pros

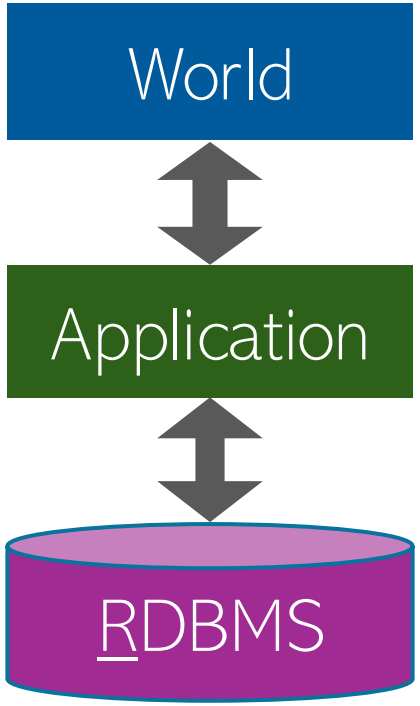
Simple SQL

Able to do very complex tasks

Cons

Slow, moving all data to application

Waste of network bandwidth



DB Level

Pros

Fast, no need to move data

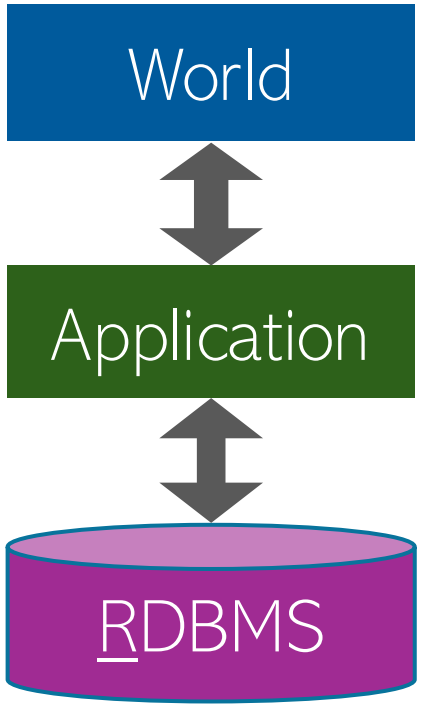
Fast, DBMS is on a powerful machine

Cons

Master SQL

Unable to do complex tasks

SQL: Ad Hoc Queries



Obtaining information as need arises,
e.g., "*which director has made the most movies?*"

Not a routine as in DAL for `INSERT`, `UPDATE`, `DELETE`, `SELECT` by Id

→ Directly to the RDBMS.